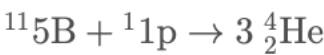


Section: Particles and Waves

Topic: Nuclear Reactions

A nuclear reaction is shown in which a proton collides with a boron nucleus, producing three alpha particles:



(a)
Show that this reaction satisfies conservation of nucleon number and atomic number

Let’s check **nucleon number** (mass number, A) and **atomic number** (Z):

- Left-hand side:
 $A = 11 + 1 = 12, \quad Z = 5 + 1 = 6$
- Right-hand side:
 $3 \times {}^4_2\text{He} \Rightarrow A = 3 \times 4 = 12, \quad Z = 3 \times 2 = 6$

✔ Both **A** and **Z** are conserved. The reaction is valid.

(b)
Calculate the energy released in this reaction

Use:
 $E = \Delta m \cdot c^2$

Step 1: Find total mass before and after

Particle	Mass (u)
${}^{11}_5\text{B}$	11.00931
${}^1_1\text{p}$	1.00728
${}^4_2\text{He}$	4.00150

- Total mass before:
 $11.00931 + 1.00728 = 12.01659 \text{ u}$
- Total mass after:
 $3 \times 4.00150 = 12.00450 \text{ u}$

$$\Delta m = 12.01659 - 12.00450 = 0.01209 \text{ u}$$

Step 2: Convert to energy using $1 \text{ u} = 931.5 \text{ MeV}$

$$E = 0.01209 \times 931.5 = 11.3 \text{ MeV}$$

🧠 Revision Tips

- Check **mass number and atomic number separately** for nuclear equations
- Always use **precise masses** from the question for energy calculations
- Energy release = **mass lost** × conversion factor (931.5 MeV/u)
- A small mass loss corresponds to a **large energy release**, showing how powerful nuclear reactions are